

- 40** The nucleus of a radioactive isotope of an element emits an α -particle. The daughter nucleus then emits a β^- particle and then the daughter nucleus of that reaction emits another β^- particle.

Which statement describes the final nuclide that is formed?

- A** It is a different isotope of the original element.
- B** It is a nuclide of a different element of higher proton number.
- C** It is a nuclide of the same element but with different proton number.
- D** It is identical to the original nuclide.